Blynt — Complete Use Cases Specification

Exhaustive Architectural Flowcharts & Business User Journeys Across All 14 System Scenarios

Document Overview: This document provides an exhaustive, visual specification of every major business and client-facing use case within the Blynt accounting platform. Each scenario features a standardized architectural flowchart following the vertical validation pattern: descending sequentially through security, business logic, and mathematical integrity checks before committing to the general ledger, with every failure condition mapped to explicit domain error branches.

UC-01: User Authentication & Multi-Tenant Access

Primary Actor: Any User (Owner, Accountant, Staff) | **Rejection Code:** 401 / 403 / TENANT_NOT_FOUND

Objective: Authenticate user credentials and securely resolve organization context without cross-tenant data leaks.

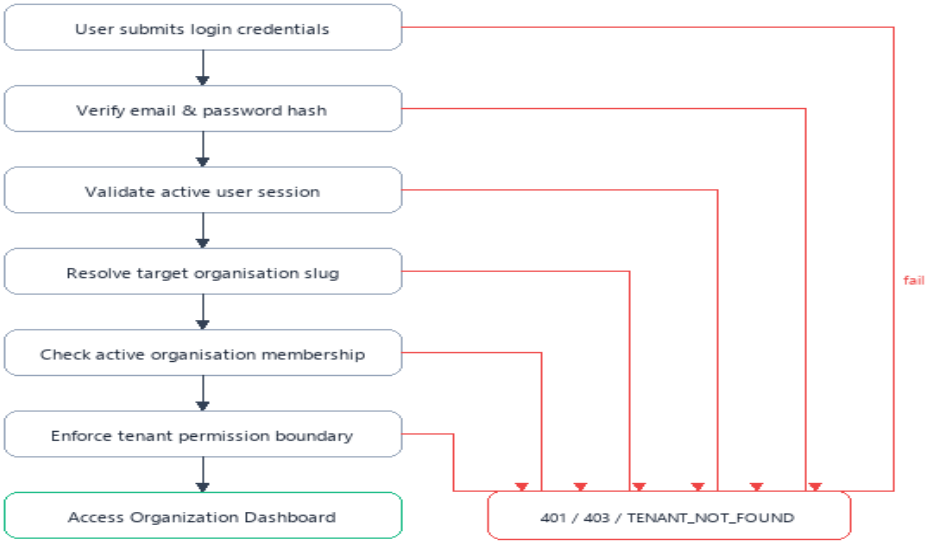


Figure UC-01 — User Authentication & Multi-Tenant Access Flow Architecture

Context: Ensures that every user strictly operates inside their authorized business entity. If an expired token, wrong password, or unassigned organization is requested, the system instantly halts access.

UC-02: Organization Setup & Statutory Tax Configuration

Primary Actor: Business Owner / Enterprise Administrator | **Rejection Code:** 400 / 409 / SLUG_OR_VAT_INVALID

Objective: Register a new UK entity, base currency (GBP), and statutory HMRC VAT number.

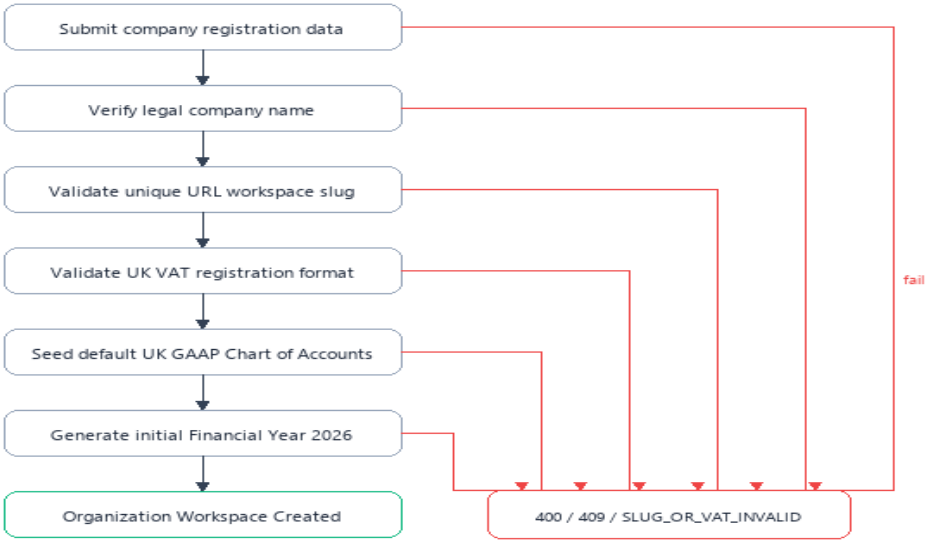


Figure UC-02 — Organization Setup & Statutory Tax Configuration Flow Architecture

Context: Validates business entity identity and seeds standardized nominal ledger accounts with VAT control accounts ready for trading.

UC-03: Chart of Accounts & Control Account Governance

Primary Actor: Certified Public Accountant | **Rejection Code:** 400 / 422 / CONTROL_PROTECTED

Objective: Create custom nominal codes while protecting system control accounts (Bank, AR, AP, VAT).

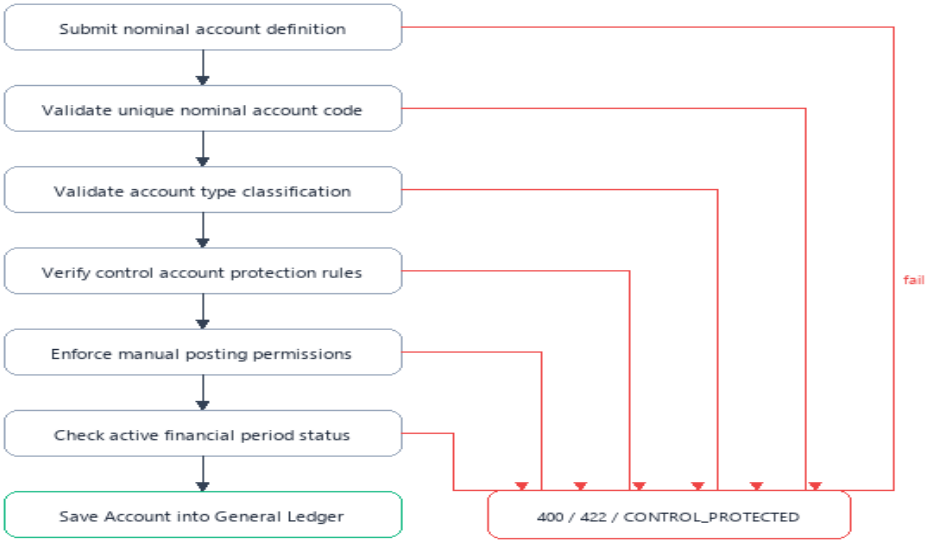


Figure UC-03 — Chart of Accounts & Control Account Governance Flow Architecture

Context: Prevents unauthorized users from manually posting direct journal lines into statutory accounts like Accounts Receivable or Tax Control.

UC-04: Sales Invoice Creation & 20% VAT Calculation

Primary Actor: Business Owner or Billing Clerk | **Rejection Code:** 400 / 422 / INVOICE_VALIDATION_ERR

Objective: Bill a customer with automatic UK standard VAT derivation and real-time subledger posting.

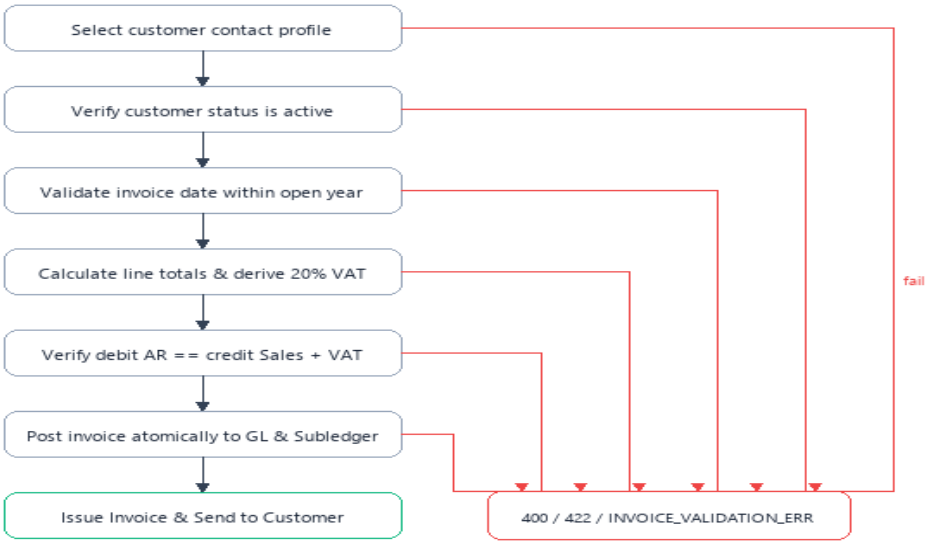


Figure UC-04 — Sales Invoice Creation & 20% VAT Calculation Flow Architecture

Context: Guarantees that invoices always balance mathematically, calculate exact VAT pence, and synchronously update customer balances.

UC-05: External Client Online Payment (Client-Side)

Primary Actor: External Customer / Payer | **Rejection Code:** 400 / 402 / CARD_DECLINED_OR_EXPIRED

Objective: Allow external clients to open secure invoice links and pay via card without creating an IT account.

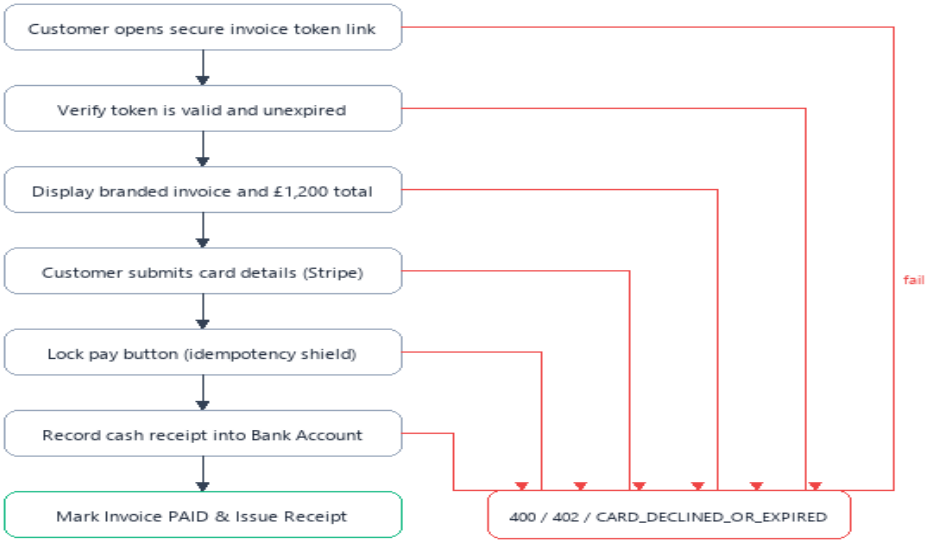


Figure UC-05 — External Client Online Payment (Client-Side) Flow Architecture

Context: Enables immediate payment receipting while safeguarding payers against duplicate charges if clicking 'Pay' rapidly.

UC-06: Supplier Bill Capture & Accounts Payable

Primary Actor: Bookkeeper / Finance Administrator | **Rejection Code:** 400 / 409 / DUPLICATE_SUPPLIER_BILL

Objective: Log vendor bills, check for duplicate invoice references, and accrue accounts payable.

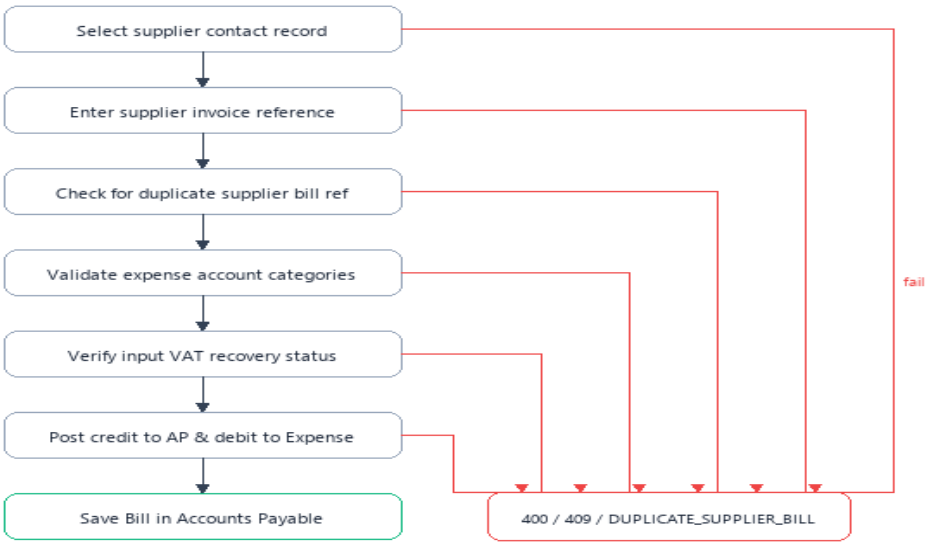


Figure UC-06 — Supplier Bill Capture & Accounts Payable Flow Architecture

Context: Prevents the business from accidentally paying the same supplier invoice twice by checking vendor numbers and dates.

UC-07: Smart AI Receipt Scanning & Document Extraction

Primary Actor: Staff Member / Expense Submitter | **Rejection Code:** 422 / LOW_CONFIDENCE_OR_UNBALANCED

Objective: Extract vendor, date, and VAT breakdown from a photo receipt using AI with zero hallucinations.

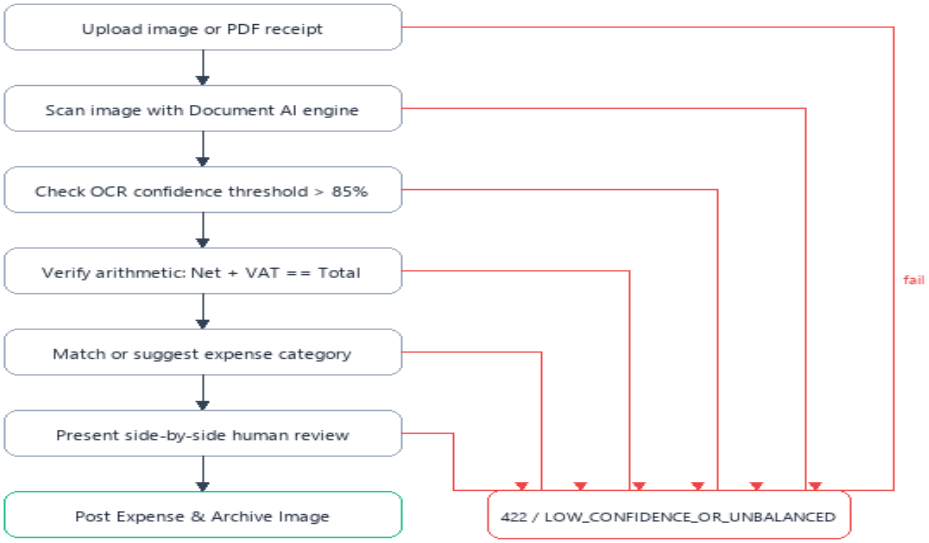


Figure UC-07 — Smart AI Receipt Scanning & Document Extraction Flow Architecture

Context: Automates receipt data entry with strict guardrails: if text is blurry or totals do not add up, human review is enforced.

UC-08: Double-Entry Manual Journal Entry Creation

Primary Actor: Senior Accountant | **Rejection Code:** 400 / 422 / JOURNAL_UNBALANCED

Objective: Post end-of-month adjusting journal entries with strict debit-equals-credit equilibrium.

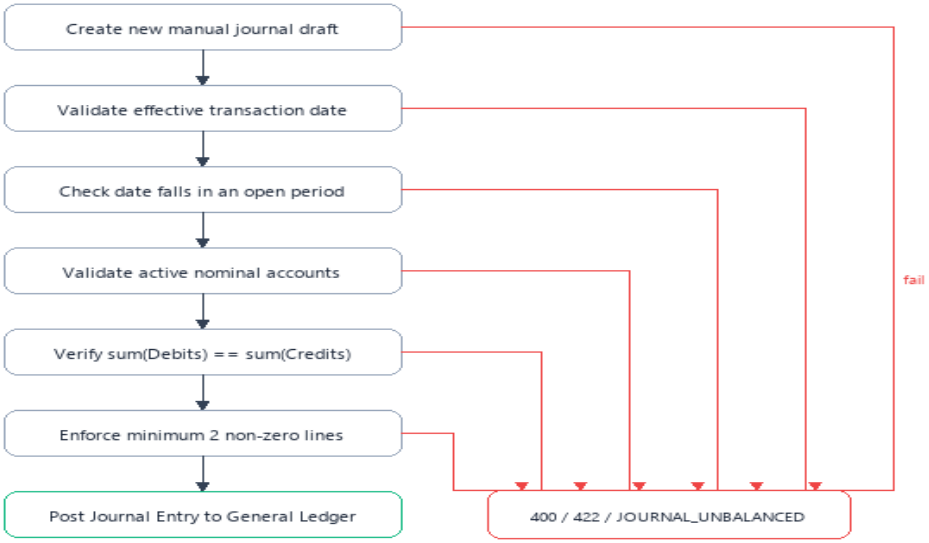


Figure UC-08 — Double-Entry Manual Journal Entry Creation Flow Architecture

Context: Enforces the foundational rule of accounting: the system physically refuses to save any journal where debits do not match credits.

UC-09: Safe Invoice Voiding & GAAP Symmetric Reversal

Primary Actor: Business Owner / Certified Accountant | **Rejection Code:** 400 / 409 / INVOICE_ALREADY_PAID

Objective: Legally cancel an invoice with an offsetting symmetrical journal reversal without erasing history.

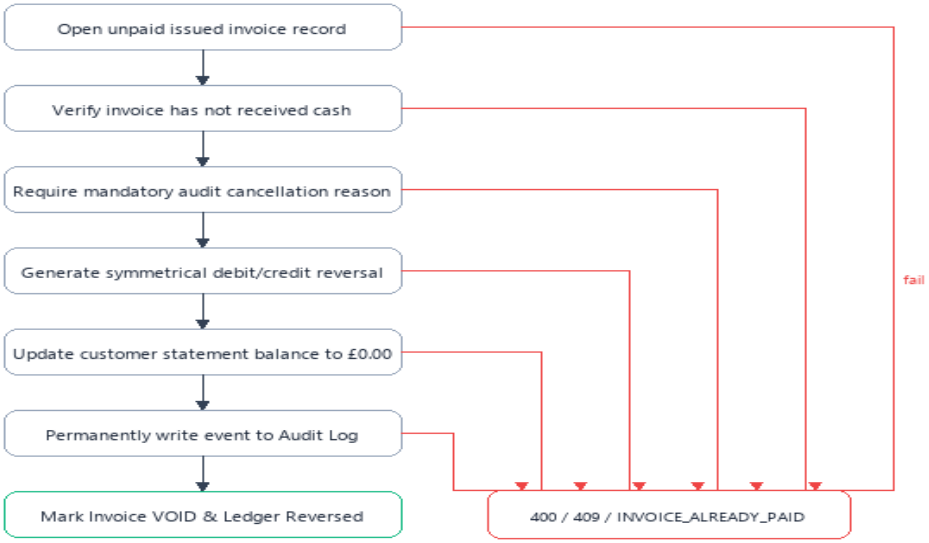


Figure UC-09 — Safe Invoice Voiding & GAAP Symmetric Reversal Flow Architecture

Context: Preserves a 100% compliant paper trail for statutory HMRC audits; issued invoices are never silently deleted.

UC-10: Financial Period Closing & Locked Year Freeze

Primary Actor: Certified Public Accountant | **Rejection Code:** 400 / 409 / UNPOSTED_DRAFTS_EXIST

Objective: Close a financial year after tax submission to freeze all past ledger entries against tampering.

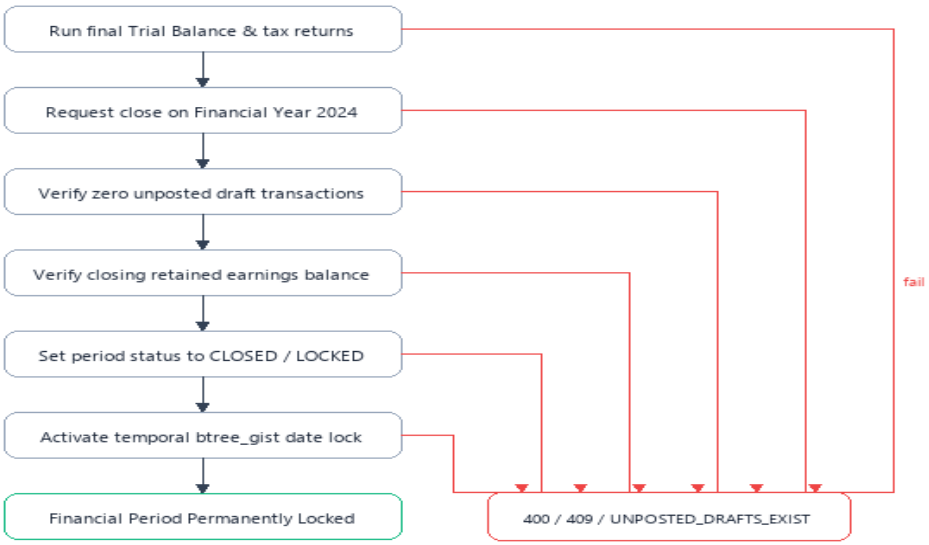


Figure UC-10 — Financial Period Closing & Locked Year Freeze Flow Architecture

Context: Once locked, any subsequent attempt by any user to post, edit, or backdate entries into that year is rejected.

UC-11: Bank Feed Statement Import & Reconciliation

Primary Actor: Bookkeeper / Accountant | **Rejection Code:** 400 / 409 / DUPLICATE_STATEMENT_FEED

Objective: Import electronic bank statement lines and reconcile against issued invoices and bills.

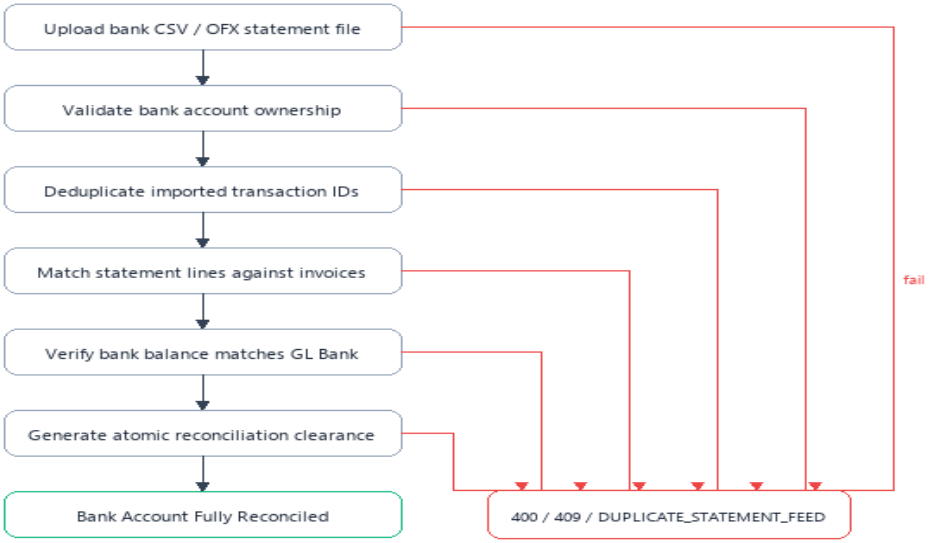


Figure UC-11 — Bank Feed Statement Import & Reconciliation Flow Architecture

Context: Matches actual bank cash flows with accounting records and catches duplicate bank transaction downloads.

UC-12: Role-Based Access Control & Privilege Boundary

Primary Actor: Platform Administrator / Business Owner | **Rejection Code:** 403 / PERMISSION_DENIED

Objective: Enforce strict least-privilege security between Owners, Accountants, Bookkeepers, and Viewers.

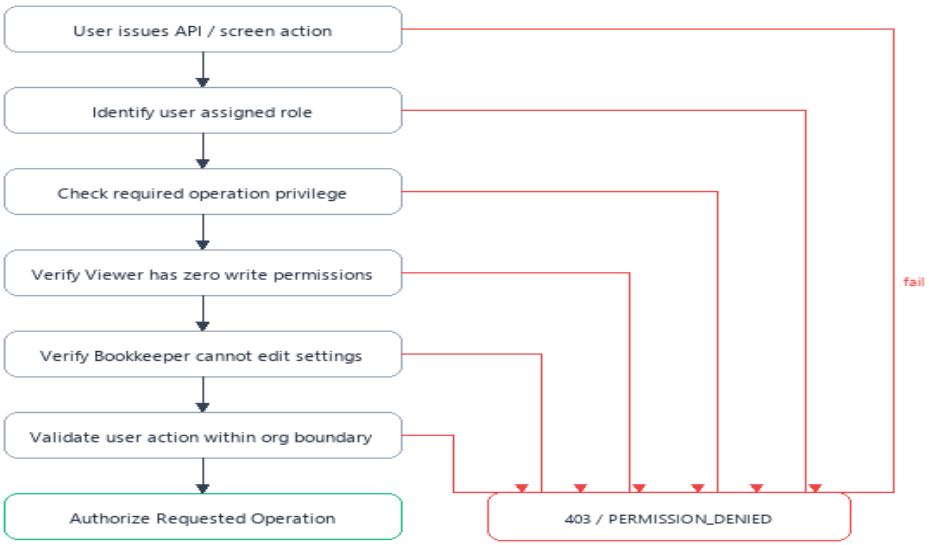


Figure UC-12 — Role-Based Access Control & Privilege Boundary Flow Architecture

Context: Ensures external auditors or junior staff cannot alter sensitive bank accounts, delete records, or modify billing configurations.

UC-13: HMRC Making Tax Digital (MTD) VAT Return

Primary Actor: Certified Public Accountant | **Rejection Code:** 400 / 422 / HMRC_API_REJECTED

Objective: Calculate quarterly VAT Boxes 1 through 9 from general ledger lines and submit to HMRC MTD.

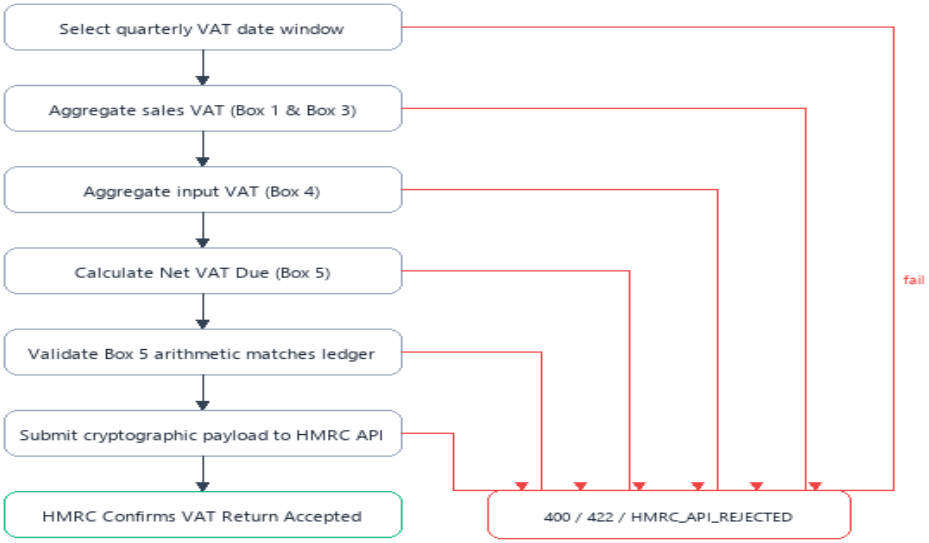


Figure UC-13 — HMRC Making Tax Digital (MTD) VAT Return Flow Architecture

Context: Generates 100% compliant UK VAT filings directly from posted invoices and receipts without spreadsheet copy-paste errors.

UC-14: Statutory Tamper-Proof Audit Trail Logging

Primary Actor: Compliance Auditor / Tax Inspector | **Rejection Code:** 403 / HASH_CHAIN_COMPROMISED

Objective: Inspect the complete historical log of all creations, edits, reversals, and access logs.

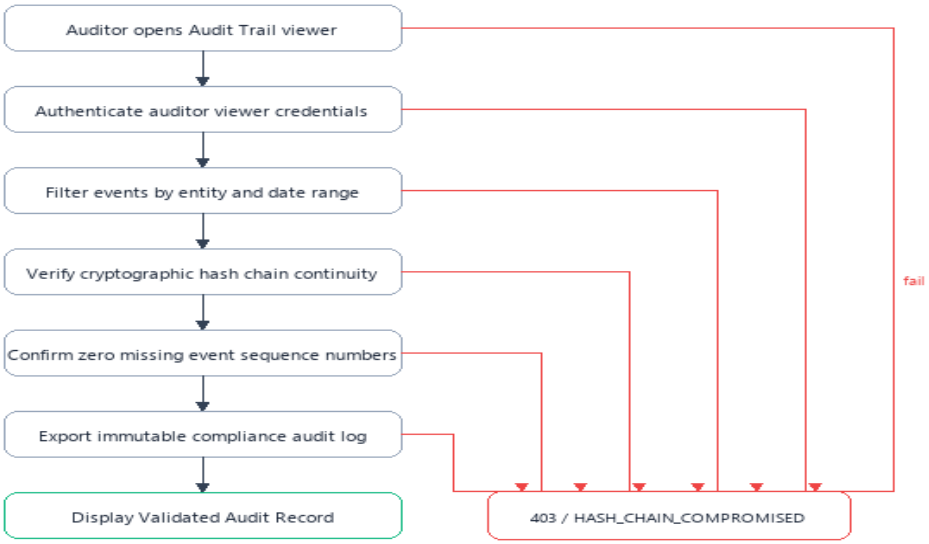


Figure UC-14 — Statutory Tamper-Proof Audit Trail Logging Flow Architecture

Context: Provides statutory proof that no accounting records were altered off-the-books, satisfying UK Companies Act requirements.